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New palaeomagnetic results from the 'cap dolomite' of the Neoproterozoic Walsh Tillite, northwestern Australia

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Abstract

An interpreted primary magnetic remanence that passed a fold test has been revealed from six sampling sites in the pink 'cap dolomite' of the Neoproterozoic Walsh Tillite, southern Kimberley, northwestern Australia. A palaeomagnetic pole (WTC) at (21.5°S, 282.4°E) with dp = 12.2° and dm = 15.4°, based on results of the five best sites, gives a palaeolatitude of $45 \pm 12°$ for the sampling region, and ca $25 \pm 12°$ for the Adelaide Fold Belt where two Neoproterozoic glaciations occur. This palaeopole falls 58° away from the Marinoan glaciation (the Elatina Formation) pole, and the palaeolatitude for the region studied is around 30° higher than that predicted by the Marinoan and younger Neoproterozoic poles from Australia. If the WTC pole is accepted as a palaeopole at the end of the Walsh glaciation, the Walsh Tillite is thus likely to be of Sturtian (ca 770–750 Ma) rather than Marinoan age. A comparison of the revised late Mesoproterozoic to Neoproterozoic apparent polar wander path (APWP) for East Gondwana with that of Laurentia suggests that supercontinent Rodinia broke apart by about 750 Ma, after the Sturtian glaciation. The new data do not discriminate between the snowball Earth model and the high-obliquity Earth model for the Neoproterozoic time. The revised APWP suggests a 'stop-and-go' fashion of movement for East Gondwana during the Neoproterozoic. Published by Elsevier Science B.V.

Keywords: Australia; Cap dolomite; Neoproterozoic; Palaeomagnetism; Rodinia; Walsh Tillite

1. Introduction

Neoproterozoic glacial deposits have attracted wide attention in recent years for two reasons. First, their occurrence is believed to have been related to supercontinent breakup (e.g. Young, 1995), and has been used as an important marker for stratigraphic correlations between different parts of those supercontinents (e.g. Eisbacher, 1985; Young, 1992; Powell et al., 1994). Second, unlike their Phanerozoic counterparts, they have often been reported as occurring close to the palaeo-equator (e.g. Embleton and Williams, 1986; Schmidt and Williams, 1995; Park, 1997). This has prompted geodynamic and palaeoclimatic models significantly different from the one we know today, such as the snowball Earth model (Kirschvink, 1992), and the high-obliquity (> 54°) Earth model (Williams, 1975). More recently, Evans (1998) invoked true polar wander (TPW) to explain possibly rapid latitudinal changes of some continents during late Neoproterozoic time. However, as pointed out by Hoffman et al. (1998), the TPW hypothesis does not explain the low palaeomag-

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netic inclinations observed directly from Neoproterozoic glacial deposits.

Key to the testing of these geodynamic/palaeoclimatic models, as well as the verification of the Proterozoic supercontinent hypothesis (e.g. Dalziel, 1991; Hoffman, 1991; Moores, 1991), is to have more reliable palaeomagnetic data from the Neoproterozoic rocks, particularly the glacial units, from widespread geographic localities (e.g. Powell et al., 1993; Meert and Van der Voo, 1994; Torsvik et al., 1995; Weil et al., 1998). In this paper I report new palaeomagnetic results from the 'cap dolomite' of the Walsh Tillite in southern Kimberley craton, northwestern Australia, and examine their significance in stratigraphic correlation, supercontinent evolution, and global geodynamic models.

2. Regional geology and palaeomagnetic sampling

The study area is in the southern Kimberley block, just north of the northwesterly trending King Leopold Orogen [Fig. 1(a)], where the Neoproterozoic succession lies unconformably over the Palaeoproterozoic Kimberley Basin succession. The Neoproterozoic succession, known as the Mount House Group, is about 500 m thick, and consists of the Walsh Tillite with pink 'cap dolomite', the Traine Formation sandstone, the Throssell Shale, and the Estaughs Formation silty sandstone [Fig. 1(b)]. There are few precise age constraints for the succession apart from an Rb–Sr whole-rock isochron age of 671 ± 72 Ma [Bofinger, 1967; recalculated in Plumb (1981)] for the Throssell Shale, and a ca 560–530 Ma K–Ar age for the deformation along the King Leopold Orogen (the Precipice Fold Belt, Tyler and Griffin, 1990; Shaw et al., 1992), that folded the succession.

The correlation between the Walsh Tillite and other glacial units in eastern Kimberley, and glacial deposits in other parts of Australia, has been debated for two decades. Whereas Dow and Gemuts (1969) and Plumb and Gemuts (1976) correlated the Walsh Tillite and the Landrigan Tillite in East Kimberley with the Sturtian glacial deposits in the Adelaide Fold Belt, more recent workers (Plumb, 1981; Grey and Corkeron, 1998) correlate them with the younger Marinoan glacial deposits. Coats and Preiss (1980) correlated the Walsh Tillite with the younger Egan glacial unit in East Kimberley, and with the Marinoan glacial deposits in the Adelaide Fold Belt.

The palaeomagnetism of the Mount House Group was previously investigated by McWilliams (1977). He sampled all the formations except the Traine Sandstone, but only 12 samples from the purple ferruginous sandstone of the Estaughs Formation revealed a stable remanent direction that is different from the present magnetic field direction. The pink dolomite at the top of the Walsh Tillite was not sampled.

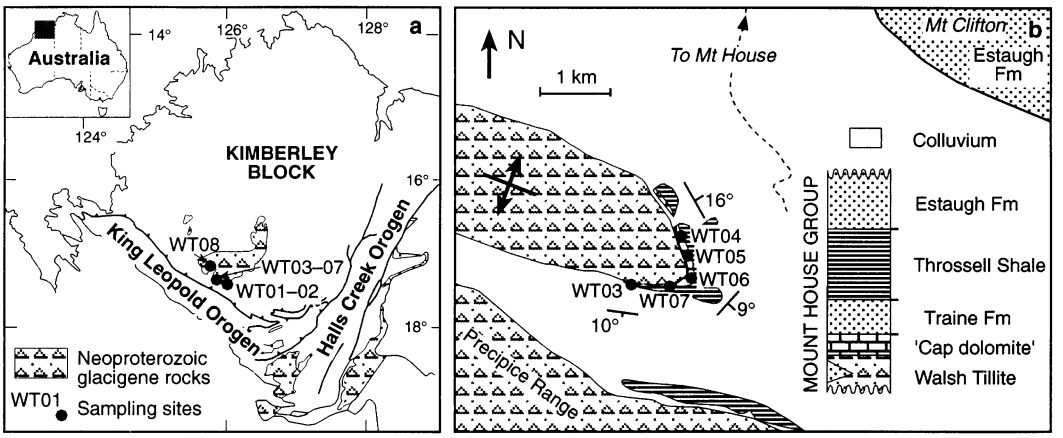


Fig. 1. Regional geology, stratigraphy and sampling localities.

A total of 64 independently oriented samples, either as blocks or 25 mm diameter drill cores, were collected from three localities during this study: sites WT01 and WT02 from the western bank of the Throssell River [Fig. 1(a)], sites WT02–WT07 from an anticline south west of Mt Clifton [Fig. 1(a) and (b)], and site WT08 from a creek cut northwest of Mt Clifton. Site WT01 was from the diamictite, and WT02 to WT08 from the 'cap dolomite' of the tillite. Bedding at sites WT01, WT02 and WT08 is close to horizontal, and at the anticlinal sites has dip angles ranging between 7 and 20°. Near-vertical axial cleavage has been observed at site WT07, but was mostly avoided during sampling. Both magnetic compass and Sun compass measurements were taken for sample orientations, and five bedding-measurements were taken for most of the sites.

3. Palaeomagnetic results

The samples were processed at the palaeomagnetism laboratory at the University of Western Australia. At least one specimen from each sample were subjected to stepwise thermal demagnetisation using MMTD2 furnaces manufactured by Magnetic Measurements, and one or two specimens from each site were subjected to stepwise alternating field (AF) demagnetisation using a 2G600 automated two-axis degaussing system. Magnetic remanence was measured using a 2G755R cryogenic magnetometer. All instruments are hosted in a three-axis Helmholtz coil system with each set measuring 3.8 m × 3.8 m. Principal component analysis (Kirschvink, 1980) and graphic display were carried out using the CSIRO Rockmag software package.

Six out of the seven sites from the pink 'cap dolomite' revealed two components: a low-temperature component (C1) parallel to the present local geomagnetic field (Figs. 2 and 3), and a high-temperature component (C2) that directs away from the present geomagnetic field direction (Figs. 2 and 4). Thermal and AF demagnetisation spectra indicate that C1 is mainly carried by very fine-grained haematite with some contribution from goethite, and is probably a combination of viscous magnetisation and weathering-related chemical remanence. C2, on the other hand, is

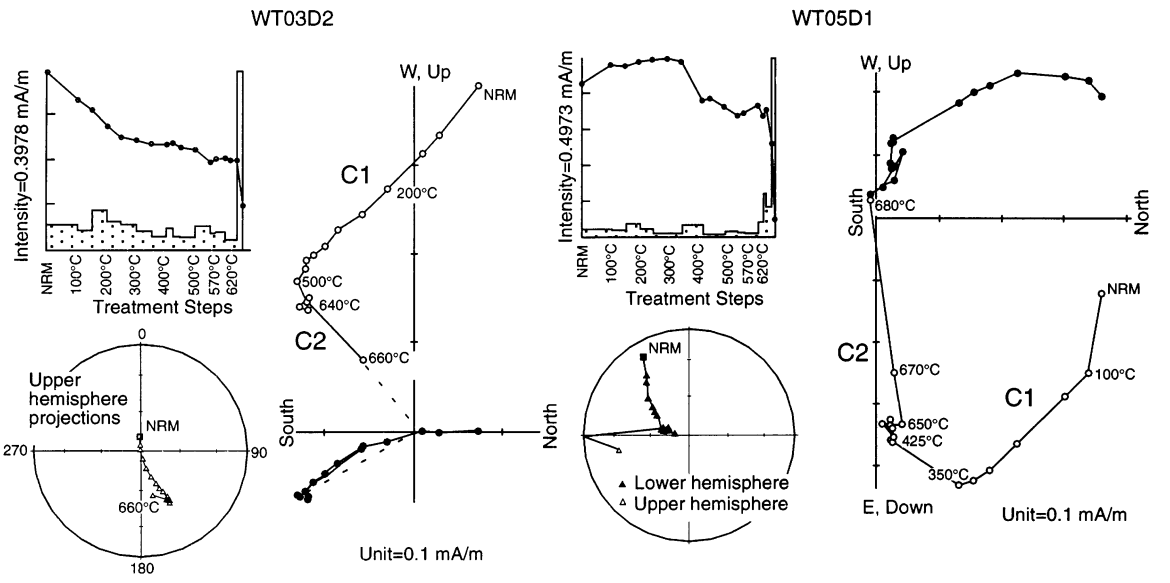


Fig. 2. Representative demagnetization results (in situ). In orthogonal plots, filled circles are projections on the horizontal plane and open circles are projections on the vertical planes. In stereographic plots, open symbols are upper hemisphere projections, and filled symbols lower hemisphere projections. C1 and C2 are the two major remanence components revealed from the 'cap dolomite'.

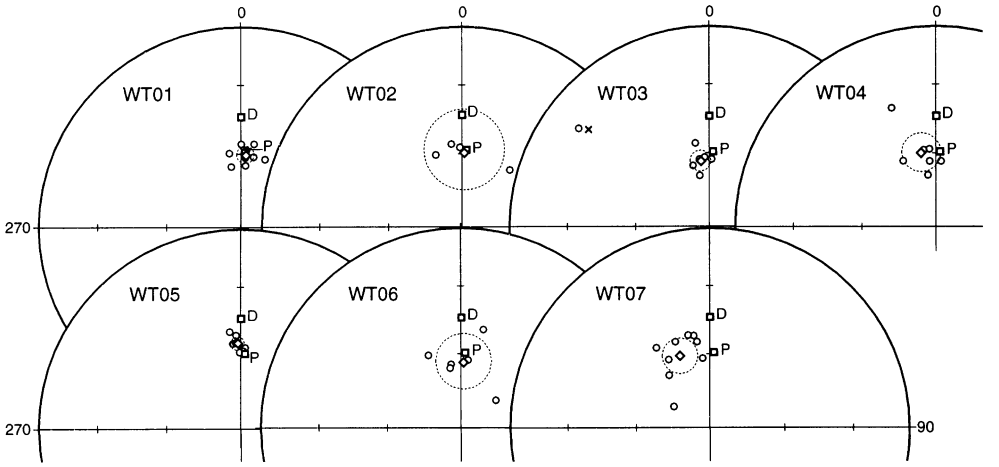


Fig. 3. Equal-angle stereographic plots of the C1 directions. Diamonds are the site-means. D: present dipole field direction; P: present local geomagnetic field direction. The direction marked with '×' in site WT03 is considered an outlier. Conventions follow Fig. 2.

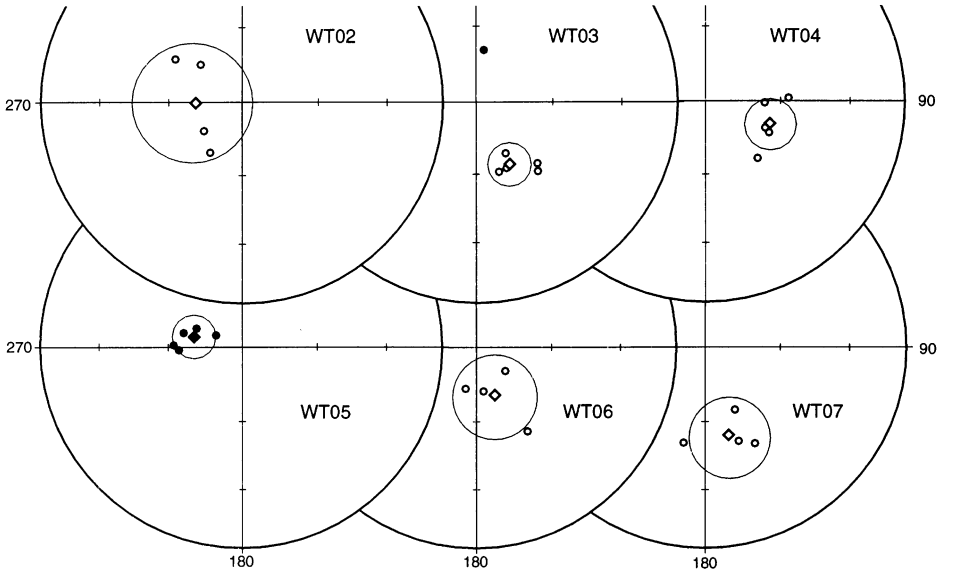


Fig. 4. Equal-angle stereographic plots of the C2 directions before structural correction. Conventions follow those in Figs. 2 and 3.

carried entirely by haematite, as suggested by its high unblocking temperature. The number of samples that have a clearly isolated C2 component was reduced to about 50% of the total sample number from the six sites due to two factors. The first one is that some specimens shattered during the 350°C heating step, possibly because of changes in their chemical/mechanical properties. The other is that the demagnetisation behaviour of some samples became erratic before the C2 component was isolated. Nonetheless, all six sites have more than four samples remaining, and the C2 Fisher (1953) distributions from five of the sites have α_{95} below 30° (Table 1 and Fig. 4). Site WT02 was rejected from further data analysis because of its large α_{95}. Sites WT01 (diamictite) and WT08

Table 1
Summary of the palaeomagnetic results from the 'cap dolomite' of the Walsh Tillite (Location 17.35°S, 125.85°E)[a]

Site	*n*/no	In situ		Structural corrected		Common		Magnetic polarity[b]
		Dec. (deg)	Inc. (deg)	Dec. (deg)	Inc. (deg)	α_{95} (deg)	*k*	
Primary remanence								
WT02[c]	4/9	269.2	−63.8	269.2	−63.8	31.3	9.6	R
WT03	6/7	153.1	−51.6	145.6	−60.2	11.1	37.3	5/6R
WT04	5/8	109.9	−52.4	130.3	−60.7	13.0	35.4	R
WT05	4/7	282.9	62.6	316.9	70.5	11.7	62.4	N
WT06	4/6	158.8	−61.4	169.5	−69.2	22.4	17.8	R
WT07	4/9	164.7	−41.2	159.9	−52.3	18.9	24.5	R
Total mean		140.6	−56.6	148.6	−63.3			
($N=5$,		$\alpha_{95}=17.9°$,		$\alpha_{95}=9.8°$,				
SCOS 95% = 2.6,		$k=19.2$		$k=62.3$				
SCOS 99% = 3.6)		SCOS1 = 2.9		SCOS1 = 2.2				
		SCOS2 = 4.0		SCOS2 = 2.1				
Palaeopole position				(21.5°S, 282.4°E)				
(South Pole WTC)				dp = 12.2°, dm = 15.4°				
Remanence of recent origin								
WT01	9/11	3.6	−50.2			4.5	131.5	N
WT02	4/9	1.4	−49.6			20.1	21.9	N
WT03	6/7	352.6	−53.0			5.4	156.1	N
WT04	7/8	348.9	−48.7			9.8	39.1	N
WT05	7/7	357.7	−43.7			3.4	315.9	N
WT06	6/6	2.3	−54.2			14.6	22.0	N
WT07	9/9	337.2	−47.7			9.1	33.3	N
Mean	$N=7$	354.7	−49.9			5.3	132.3	
South Pole position at (75.8°S, 144.9°E), dp = 4.7°, dm = 7.1°								

[a] *n*/no: number of samples revealing the characteristic remanence/total number of samples; *N*: number of sites; α_{95}: half-angle of the 95% confidence cone for mean direction; Dec.: declination; Inc.: inclination; *k*: precision parameter of Fisher (1953) distribution for mean directions; dp, dm: semi-axes of elliptical error around the pole at a probability of 95%; SCOS: critical value of McFadden (1990) correlation test statistics, where '2' stands for definition two.

[b] N: normal; R: reversed.

[c] Site excluded from mean calculation because of large α_{95}.

(leached 'cap dolomite') failed to reveal the characteristic component.

Four lines of evidence suggest a possibly primary origin for the C2 component.

First, a positive fold test. As shown in Fig. 5, the five site-mean directions group better after bedding correction (*k* increases from 19.2 to 62.3 and α_{95} reduces from 17.9° to 9.8°). A McFadden (1990) correlation fold test (Table 1) indicates that the null hypothesis of magnetisation being acquired after folding can be rejected with 95% confidence using his test definition one, and with 99% confidence using test definition two. However, the null hypothesis of magnetisation being acquired at stratigraphic coordinates could not be rejected using either test definition. It is thus argued that the C2 component was acquired before the ca 560–530 Ma deformation. If the Walsh Tillite is of Sturtian age (Dow and Gemuts, 1969; Plumb and Gemuts, 1976) (see more discussion in Section 4.1), the magnetic remanence age is constrained to be within ca 200 My after the rock formation. However, if the tillite is Marinoan age or younger, the fold test would constrain the age of the remanence to be within ca 50–100 My after deposition. The bedding-corrected mean direction of $D=148.6°$ and $I=-63.3°$ ($N=5$ sites, $\alpha_{95}=9.8°$) gives a palaeopole (WTC) position of (21.5°S, 282.4°E) with dp = 12.2° and dm = 15.4°. Because the sampling sites are located around the

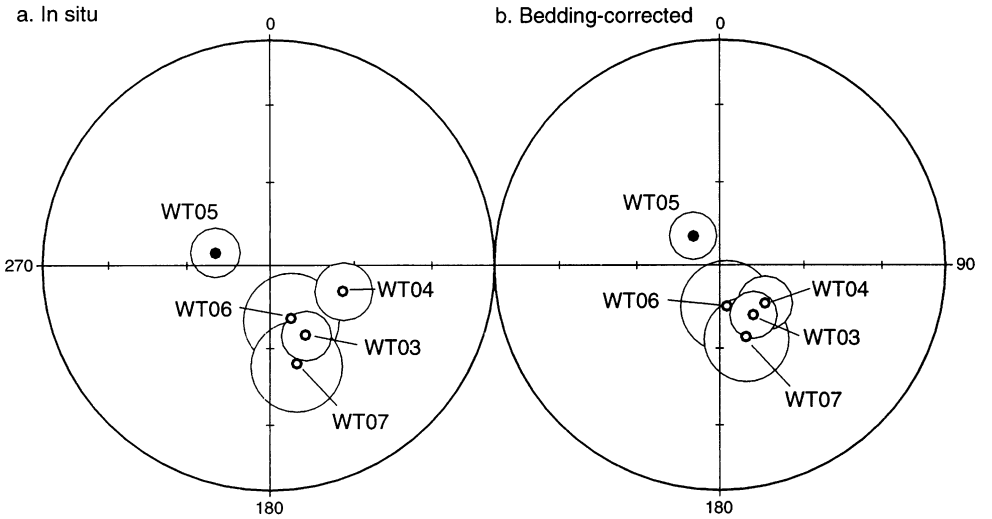


Fig. 5. Effect of structural correction on the site-mean directions of C2. Conventions same as in Fig. 2.

nose of a near-horizontal anticline with very gentle dips [Fig. 1(b)], a simple bedding-correction has been applied to all of them.

Second, the dissimilarity between palaeopole WTC and known late Neoproterozoic palaeopoles for Australia. Given that there is no geological evidence suggesting large-scale relative movement between the Kimberley craton and the rest of the Australian craton since the Neoproterozoic, the WTC pole is compared with other Neoproterozoic palaeopoles from Australia. As shown in Fig. 6, the WTC pole falls away from all the Neoproterozoic poles except for the poorly dated YB pole (Table 2). It can thus be argued that the pre-folding result is unlikely to be a late Neoproterozoic overprint.

Third, the lack of any pervasive magnetic overprint in the Kimberley craton as shown by the study of both the 1790 Ma (R.W. Page, in Blake, 1997) Hart Dolerite sills (McElhinny and Evans, 1976), which are widespread in the Kimberley block, and the ca 1800 Ma Elgee Formation red siltstones (Li, in preparation).

Fourth, because the C2 component is carried by haematite, which is shown in thin-sections as red pigment in bedding-parallel wavy silt/mud laminae, it is likely to be either depositional or early diagenetic origin. The haematite gives the dolomite the pink colour.

The presence of dual polarities and the fact that the remanence is carried by sedimentary rocks

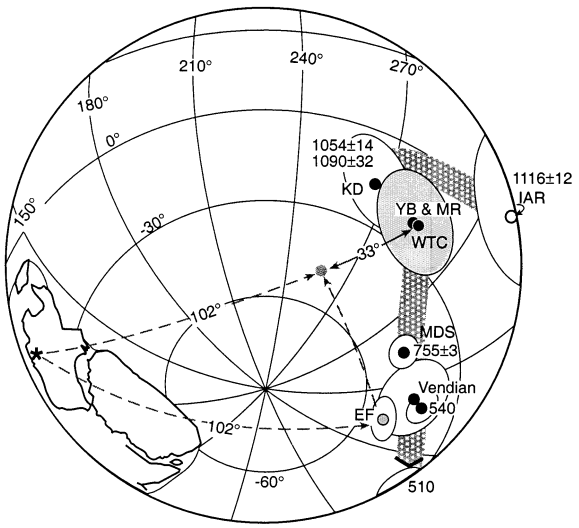


Fig. 6. Apparent polar wander path of East Gondwana for the ca 1100–510 Ma interval in present Australian coordinates. Poles used and their sources are given in Table 2. Pole WTC (with shaded A_{95}) is the new pole documented in this study. Schmidt equal area projection with centre at (50°S, 230°E).

Table 2
List of palaeomagnetic poles used in Figs. 6 and 7

Pole/Age (Ma)	Lat. (°N)	Long. (°E)	dp/A_{95}	dm	N[a]	Sources[b]
East Gondwana (in present Australian coordinates)						
Early to mid-Camb. (ca 510)	−31.8	12.1	6.0		8	Aus: HKG, AS, KI, BWA and LFG2; Ind: BS1, PS (purple sandstone; McElhinny, 1970), and SPB [salt pseudomorph beds, after Wensink (1972)].
ca 540	−40.8	342.2	4.8		4	Aus: TAE and AD (Albany dolerite intrusives; Harris and Li, 1995); Ind: BS2 and BS3
Vendian (ca 600)	−44.3	337.8	11.5		4	Aus: EF (Elatina Formation; Schmidt and Williams, 1995), BF, UAS and PA
(EF, ca 600	−52.4	347.1	3.7	7.4)		
MDS (755±3)	−45.8	315.3	4.3		1	Aus: Combined NDD and Mundine Well dyke swarm data (Wingate, 1997; Wingate and Giddings, 1999)
YB+MR	−20.8	284.3			2	Aus: YB; Ind: MR
WTC	−21.5	282.4	12.2	15.4	1	Aus: "cap dolomite" of Walsh Tillite (this study)
KD (1054±14; 1090±32)	−17	266	12		1	Aus: Kulgera dike swarm (Camacho et al., 1991)
IAR (1116±12)	9.5	311.1	17.4		1	Aus: Mt Isa dykes, Tanaka and Idnurm (1994); Duff and Embleton (1976)
Laurentia (in present North American coordinates)						
Vendian (ca 580)	43.5	304.5	7.6		3	See table 2 of Powell et al. (1993)
ca 725	−5.9	342.4			2	See table 2 of Powell et al. (1993)
ca 740	−22.5	341.3	8.8		3	See table 2 of Powell et al. (1993)
R2 (Rapitan Glacia.)	−1	334	4	9	1	Park (1997)
ca 780	5.5	318.0	6.1		4	See table 2 of Powell et al. (1993)
ca 810	11.0	324.0			2	See table 2 of Powell et al. (1993)
ca 830	−10.5	328.0			2	See table 2 of Powell et al. (1993)
ca 980	31.5	321.9	7.1		9	Mean of poles with assigned ages of 975 Ma and 960 Ma in table 1 of Weil et al. (1998), and Grenville poles R, N, GFTZ, K, Mu, Sh and GF of Alvarez and Dunlop (1998)
ca 1010	4.0	0.0			2	Mean of poles from the Freda and Jacobsville sandstones as listed in table 1 of Weil et al. (1998)
ca 1020	−18	340.5	15.2		4	Mean of poles with assigned ages between 1046 and 1020 Ma (excluding the Freda sandstones) in table 1 of Weil et al. (1998)
ca 1070	−27.8	353.9	8.0		4	Mean of poles with assigned ages between 1085 and 1060 Ma in table 1 of Weil et al. (1998)
ca 1100	−38.8	6.6	9.7		7	Mean of poles with assigned ages between 1113 and 1090 Ma in table 1 of Weil et al. (1998)

[a] Number of poles.
[b] Except those given in the table, sources for other poles from Australia and India can be found in table 2 of Li and Powell (1993). Aus: Australia; Ind: India.

suggest that geomagnetic secular variation is likely to have been averaged out. The C2 direction is almost identical to the stable remanence direction obtained from the flat-lying Estaughs Formation by McWilliams (1977) with $D=147°$, $I=-62°$ ($N=5$, $R=4.876$, $k=32.3$; accepted sample number: 12). The corresponding pole position for the Estaughs Formation falls at (23°S, 280°E) with dp=17°, and dm=21°. The similarity between results from the two formations suggests that both

the geomagnetic field and the position of Australia were relatively stable immediately after the Walsh glaciation.

4. Discussion

4.1. A Sturtian age for the Walsh Tillite?

As discussed earlier, it is uncertain whether the Walsh Tillite should be correlated with the Sturtian glacial deposits, or with the younger (ca 600 Ma) Marinoan glacial deposits in the Adelaide Fold Belt in South Australia. However, as shown in Fig. 6, the position of the WTC pole from the 'cap dolomite' of the Walsh Tillite, referred to here as representing the end of the glacial interval, falls 58° away from the recently revised, high-quality Marinoan pole from the Elatina Formation of the Adelaide Geosyncline (Schmidt and Williams, 1995, and previous work cited therein). This large angular difference cannot be reconciled by local vertical-axis rotation in this study region because: (1) there is a $33 \pm 16^{\circ}$ latitudinal difference in the palaeolatitude predicted for the study region by the two poles (Fig. 6); (2) the study region is structurally simple, with fold axes subparallel to the northwesterly trending King Leopold Orogen (Fig. 1). The 'cap dolomite', and by inference the conformably underlying Walsh Tillite, therefore, could not be of Marinoan age. On the other hand, if we accept the shortest apparent polar wander path (APWP) for East Gondwana as shown in Fig. 6, a more logical interpretation is that the Walsh Tillite is older than the Marinoan glaciation, and most likely of Sturtian age.

A third possibility for the age of the Walsh Tillite is that it correlates with the Egan glacial unit in southeastern Kimberley (Coats and Preiss, 1980), which has recently been interpreted by Grey and Corkeron (1998) as a second Marinoan glaciation based on similar stromatolites found in both the Egan Formation and the Julie Formation in the Amadeus Basin, Central Australia. However, because the WTC pole is about 50° away from the average Vendian pole position (Fig. 6 and Table 2), which represents palaeopoles obtained from rocks both underlying and overlying the Julie Formation in the Amadeus Basin (i.e. poles from the upper Pertatataka and Arumbera Formations, Kirschvink, 1978a,b), the Walsh Tillite is also an unlikely correlative of the putative second Marinoan glaciation.

No precise numerical age is directly available for the Sturtian glaciation. In the Adelaide Fold Belt, the age of the Sturtian glacial deposits is bounded by a 802 ± 10 Ma age from the Rook Tuff (U–Pb zircon, Fanning et al., 1986) and a 777 ± 7 Ma age (U–Pb zircon, M. Fanning, personal communication, 1998) from the Boucaut Volcanics in the basal Burra Group, both below the glacial unit (Preiss, 1993), and a 750 ± 53 Ma age (Rb–Sr) from the Tapley Hill Formation above the unit (Preiss, 1987). It should be noted that the 750 ± 53 Ma Rb–Sr age is likely to be a maximum age for the Tapley Hill Formation because of possible contamination by detrital clay minerals. The Rapitan glacial deposits from western Laurentia, which have been correlated with the Sturtian glacial deposits (e.g. Young, 1992; Powell et al., 1994), has its maximum age bounded by a 778 ± 3 Ma (U–Pb) age from diabase dykes and sills that intruded the lower Mackenzie Mountains Supergroup (Jefferson and Parrish, 1989; LeCheminant and Heaman, 1994), and a granite clast from the Rapitan glacial unit gave a U–Pb age of 755 ± 18 Ma (Ross and Villeneuve, 1997). However, its upper age limit is poorly constrained. A $751 + 26/-18$ Ma (U–Pb) age was obtained from the Mt Harper volcanic complex (Roots and Parrish, 1988), which Rainbird et al. (1996) considered no older than the Rapitan glacial units. The maximum possible age for the Sturtian–Rapitan glacial deposits is thus ca 770 Ma, but it could be as young as 700 Ma (e.g. Powell et al., 1994; Young, 1995; Veevers et al., 1997). However, when considering the relative positions between the WTC pole and the 755 ± 3 Ma MDS pole (Wingate, 1997; Wingate and Giddings, 1999) in the East Gondwana APWP (Fig. 6), the 'cap dolomite', and by inference the Sturtian–Rapitan glaciations, would appear to be closer to the older end of the time range (ca 770–750 Ma?). A similar older age for the Sturtian–Rapitan glaciations was

also suggested by Wingate and Giddings (1999) on slightly different grounds.

Such an interpretation also suggests slightly older ages for the YB pole from the Yilgarn Block dykes (Giddings, 1976) and the MR pole the Malani rhyolites of India (Klootwijk, 1975) (shown as an averaged YB and MR pole in Fig. 6), previously poorly constrained as between 700 and 750 Ma [for more detailed discussion see Wingate et al. (1998)].

4.2. Rodinia breakup between 770 and 750 Ma

If we accept a Sturtian age for the Walsh Tillite, then results of this study, for the first time, enable a direct palaeomagnetic comparison between data from the Sturtian and Rapitan glacial deposits. Fig. 7(a) plots the revised late Mesoproterozoic to Neoproterozoic APWPs for both East Gondwana and Laurentia (Table 2), where Laurentia and its palaeomagnetic poles were rotated to East Gondwana (at present Australian coordinates) using an Euler pole of (35, 135, −127) (after Powell et al., 1993). Sources for the averaged pole positions are given in Table 2. Taking into account their error ranges, the WTC pole agrees well with the R2 pole from the Rapitan glacial units (Park, 1997). The 1100–1000 Ma poles for Laurentia and East Gondwanaland (Australia) also agree with each other in this configuration. The relative positions between East Gondwana and Laurentia, as shown in Fig. 7, are thus still palaeomagnetically plausible for the 1100 Ma to 770–750 Ma interval. However, the two APWPs clearly diverge after the Sturtian–Rapitan glaciations, suggesting separations between them. This is consistent with conclusions drawn by Powell et al. (1994) based on tectonostratigraphic analysis. As pointed out by

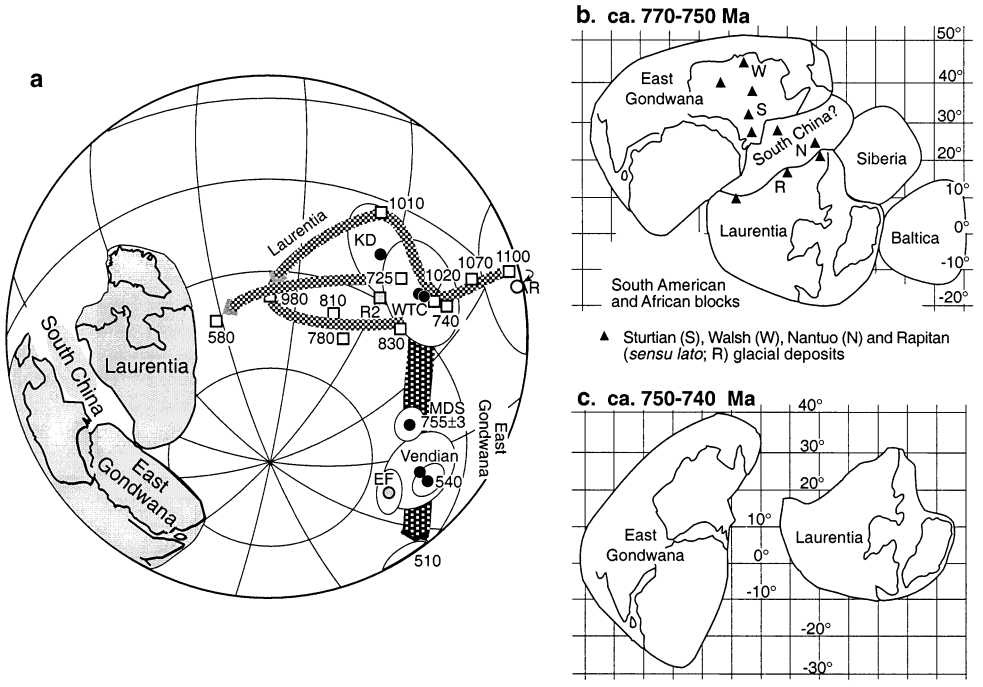


Fig. 7. (a) Late Mesoproterozoic to Neoproterozoic APWP of East Gondwana (in present Australian coordinates) and Laurentia after rotating Laurentia and its poles (Table 2) by 127° clockwise around (35°N, 135°E) (after Powell et al., 1993). South China is schematically placed between Australia and Laurentia after Li et al. (1995). A_{95} of the Laurentian poles were not shown for the sake of clarity. Map centre at (50°S, 230°E). Palaeolatitudinal positions of East Gondwana and Laurentia before (b) and after (c) the Rodinia breakup are given. Pole WTC was used in (b), and poles MDS (for East Gondwana) and 740 (Ma) (for Laurentia) were used in (c).

Wingate and Giddings (1999), the divergence of the 755±3 Ma MDS pole from the 780 to 740 Ma segment of the Laurentian APWP provides an upper age limit for the breakup, which is over 30 My earlier than the breakup age suggested by Powell et al. (1993) based on the then-available data.

4.3. A 'stop-and-go' feature of lithospheric movement?

It is interesting to note that the East Gondwana APWP leapt by about 37° between the Walsh/Sturtian glaciation time (ca 770–750? Ma) and 755±3 Ma, but appears to have moved little from 755±3 Ma until at least the mid-Vendian (the ca 600 Ma pole used is the average of four poles including the revised EF pole from the Elatina Formation). This pattern of APWP movement does not seem to be matched by that of the Laurentia APWP, which moved little until after ca 725 Ma (Fig. 7a). This could mean that: (1) there was no significant true polar wandering between the Sturtian and Vendian times; (2) East Gondwana was 'mobilised' more quickly than Laurentia (and all the other continents attached to it) in the mantle framework during the Rodinia breakup, possibly because of its smaller mass. Such a 'stop-and-go' feature of the East Gondwana, as implied by its APWP, could be a common one in the movement of all the major continents. It may reflect the rhythm of mantle convection, as modelled by the 'third regime' in Moresi and Solomatov (1998) that is characterised by episodic overturns of an unstable stagnant lithospheric lid.

4.4. Mid- to low-latitude Sturtian glaciations in Australia

The new WTC pole, interpreted as representing the palaeopole at the end of the Sturtian glaciation, gives a palaeolatitude of 45±12°N for southern Kimberley. The palaeopole predicts a palaeolatitude of ca 25±12°N for the Adelaide Fold Belt where the Sturtian glaciation was well recorded, and an equatorial position for the Rapitan glaciation in the pre-breakup configuration [Fig. 7(b)], which is consistent with the new Laurentian results (Park, 1997).

Both the snowball-Earth model (Kirschvink, 1992) and the high-obliquity ($\epsilon \geq 54°$) Earth model (Williams, 1975) explain low-latitude glaciations; however, the oblique Earth model suggests that with high Earth obliquity ($\epsilon \geq 54°$), the preferred latitudes of continental-scale glaciation should be $\leq 43°$, although weakened climatic zonation could also allow local glaciation at higher latitude (Williams, 1975). The new pole from the 'cap dolomite' of the Walsh Tillite gives a palaeolatitude of 45±12° with 95% confidence. Even if this palaeolatitude does reflect the local palaeolatitude at the time of the glaciation, it is not high enough to discriminate between the two models. Unfortunately, since it appears that the entire Rodinia supercontinent was in a mid- to low-latitudinal position at the time of the Sturtian–Rapitan glaciation [Fig. 7(b)], it would be a daunting task to search for any record of higher-latitude glaciation at ca 770–750 Ma than that determined by the WTC pole. Hence, the Marinoan–Vendian glacial interval might be more appropriate for testing these models.

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